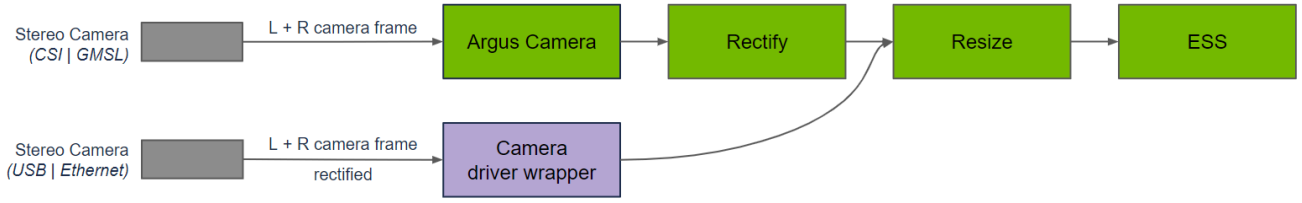
Isaac ROS DNN Stereo Depth

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Isaac ROS DNN Stereo Depth official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_dnn_stereo_depth/index.html

Overview

DNN Stereo Depth uses deep learning to estimate disparity/depth from stereo images, which provides distance information for robot manipulation, navigation and obstacle avoidance. ESS is also the stereo depth component of the Isaac ROS perception solution.



Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: all the terminals below are executed inside the container:

Terminal A: start the Nano pool=8 ESS and specify the model size that matches the 1920×1080 input:

```
ros2 launch /workspaces/isaac_ros-dev/launch/ess_nano_pool8.launch.py \
  engine_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.1.0_onnx_tr
t10.16/ess.engine \
  ess_plugin_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.1.0_onnx_tr
t10.16/plugins/aarch64/ess_plugins.so \
  input_image_width:=1920 input_image_height:=1080 \
  model_input_width:=960 model_input_height:=576 threshold:=0.0
```

Terminal B: play the ESS input bag in a loop:

```
ros2 bag play /workspaces/isaac_ros-dev/isaac_ros_assets/isaac_ros_ess/rosbags/ess_rosbag \
  --loop --rate 0.25 --remap \
  /left/camera_info:=/left/camera_info_rect \
  /right/camera_info:=/right/camera_info_rect
```

Terminal C: start the official DNN Stereo visualizer:

```
ros2 run isaac_ros_dnn_stereo_decoder isaac_ros_dnn_stereo_visualizer.py
```



When the threshold is set to 0.4, the result is shown as follows:



Terminal D: check the disparity output briefly:

```
timeout 8 ros2 topic hz /disparity
```